

The Skill Shop

Empower Your Skills

DSA + Problem Solving **18-Day Intensive Curriculum**

Campus Recruitment Training Program

Prepared for: LBRCE — Laki Reddy Bali Reddy College of Engineering

Prepared by: The Skill Shop

Duration: 18 Days | Batch: B.Tech Students

Day 1 — Bit Manipulation

Instructor Note: All conversions must avoid Python's built-in `bin()/int()`. Focus on mastering `&`, `|`, `^`, `~`, `<<`, `>>` as computational tools.

Classwork Problems

#	Problem	Diff
1	Convert a given decimal number to its binary equivalent without using <code>bin()</code> .	Easy
2	Convert a given binary number (as string) to its decimal equivalent without using <code>int()</code> .	Easy
3	Detect whether two given integers have opposite signs without using conditional operators.	Easy
4	Add 1 to a given integer using only bit manipulation — no arithmetic operators allowed.	Medium
5	Swap two integers without a temporary variable using XOR.	Medium
6	Given a number N and position K, turn the K-th bit ON; also write a variant to turn it OFF.	Medium
7	Given a number, check whether its binary representation is a palindrome.	Medium
8	Check whether a given number is a power of 2 using a single bitwise expression.	Medium
9	Count the number of set bits in a number using Brian Kernighan's Algorithm (<code>n &= n-1</code>).	Hard
10	Check whether a given number is a power of 4 using bit manipulation — no loops or recursion.	Hard

Homework Problems

#	Homework Problem	Diff
H W 1	Find the single non-duplicate in an array where every other element appears exactly twice, using XOR.	Easy
H W 2	Find the number of bits that must be flipped to convert integer A to integer B.	Medium
H W 3	Reverse the bits of a given 32-bit unsigned integer.	Medium
H W 4	Find the missing number in array <code>[0..n]</code> using XOR (no sum formula).	Medium
H W 5	Swap all odd-positioned and even-positioned bits of a 32-bit integer using bit manipulation.	Hard

Day 2 — Number-Based Problems

Instructor Note: Focus on digit-level decomposition: sum, product, factorial, and square checks. Explore each number type independently before combining.

Classwork Problems

#	Problem	Diff
1	Spy Number: check if the sum of digits equals the product of digits for a given number. Print all Spy numbers up to N.	Easy
2	Magic Number: a number is Magic if recursively summing its digits yields 1. Check if N is a Magic number.	Easy
3	Happy Number: repeatedly replace N with the sum of squares of its digits. Detect if N eventually reaches 1.	Easy
4	Perfect Number: a number equals the sum of its proper divisors. Find all Perfect numbers up to N.	Medium
5	Perfect Square: check if a number is a perfect square without using sqrt().	Medium
6	Neon Number: the sum of digits of N-squared equals N itself (e.g. 9: $9^2=81$, $8+1=9$). Find all Neon numbers in [1, 1000].	Medium
7	Buzz Number: a number divisible by 7 or ending in 7. Print all Buzz numbers in a given range [L, R].	Medium
8	Tribonacci Series: generate first N terms where $T(n) = T(n-1)+T(n-2)+T(n-3)$. Also generate the Padovan and Jacobsthal series.	Medium
9	Krishnamurthi / Strong Number: sum of factorials of digits equals the number. Find all Strong numbers up to N.	Hard
10	Evil Number: numbers whose binary has an even number of 1-bits. In range [L,R] count how many are Evil vs Odious. Also identify Strobo-strategic numbers in the same range.	Hard

Homework Problems

#	Homework Problem	Diff
H W 1	Pronic Number: product of two consecutive integers. Find all Pronic numbers in [1, N].	Easy
H W 2	Unique Number: no digit repeats. Print all unique 3-digit numbers (100-999).	Easy
H W 3	Nelson Number (111): check if N is 111 or a multiple pattern. Generalise: find all repunit numbers up to N.	Medium
H W 4	Peterson Number: equals the sum of factorials of its digits raised to the power of their 1-based position. Check if N is a Peterson number.	Medium
H W 5	Given a range [L, R], find all numbers that are simultaneously a Perfect Square AND a Perfect number.	Hard

Day 3 — Pattern-Based Problems

Instructor Note: Identify the row-column formula before coding each pattern. Aim to write any pattern in under 5 minutes after two rounds of practice.

Classwork Problems

#	Problem	Diff
1	Print a right-angle triangle of stars (*) with N rows (left-aligned).	Easy
2	Print an inverted right-angle triangle of stars with N rows.	Easy
3	Print a centred pyramid of stars with N rows (isocles triangle).	Easy
4	Print a diamond of stars (top pyramid + inverted pyramid, 2N-1 rows).	Medium
5	Print Floyd's Triangle — consecutive natural numbers arranged row-wise.	Medium
6	Print Pascal's Triangle up to N rows.	Medium
7	Print a hollow rectangle of stars (R rows x C columns — only border stars).	Medium
8	Print a spiral / snail-fill matrix of size N x N filled with numbers 1 to N ² .	Medium
9	Print a zig-zag diagonal pattern on an N x N matrix where diagonals alternate direction.	Hard
10	Print an hourglass number pattern — top triangle of row-numbers, then inverted, separated by a blank row.	Hard

Homework Problems

#	Homework Problem	Diff
H W 1	Print a hollow diamond of stars.	Easy
H W 2	Print a number pyramid where each row repeats the row number (row 3 → 3 3 3).	Easy
H W 3	Print a butterfly pattern of stars with N rows on top and N rows on bottom.	Medium
H W 4	Print a star pattern forming the letter X on an N x N grid.	Medium
H W 5	Print a 90-degree clockwise rotated right-angle triangle of ascending numbers.	Hard

Day 4 — Array Problems

Instructor Note: Key techniques: Two Pointers, Sliding Window, Prefix Sum, in-place manipulation. Aim for $O(n)$ or $O(n \log n)$ solutions everywhere.

Classwork Problems

#	Problem	Diff
1	Find the maximum and minimum element in an array in a single traversal.	Easy
2	Check if an array is sorted in ascending order in $O(n)$ time.	Easy
3	Rotate an array to the right by K positions in-place.	Easy
4	Find all pairs in an array whose sum equals a given target (Two-Sum, all pairs).	Medium
5	Move all zeros to the end of an array while maintaining the relative order of non-zero elements.	Medium
6	Find the longest subarray with sum equal to K (handles positive and negative integers using prefix-sum map).	Medium
7	Find the maximum product subarray.	Medium
8	Find the majority element (appears more than $n/2$ times) using Boyer-Moore Voting Algorithm.	Medium
9	Merge two sorted arrays without extra space in $O((m+n) \log(m+n))$ using the gap method.	Hard
10	Find the next permutation of a given array lexicographically.	Hard

Homework Problems

#	Homework Problem	Diff
H W 1	Find the second largest element in an array without sorting.	Easy
H W 2	Remove duplicates from a sorted array in-place and return the new length.	Easy
H W 3	Find the contiguous subarray with the largest sum (Kadane's Algorithm).	Medium
H W 4	Rearrange an integer array so positives and negatives alternate without extra space.	Medium
H W 5	Merge overlapping intervals from a list of [start, end] pairs.	Hard

Day 5 — String-Based Problems

Instructor Note: Core techniques: Frequency map (hashing), Two Pointers, Sliding Window, KMP. Prefer $O(n)$ approaches; brute force only as first step.

Classwork Problems

#	Problem	Diff
1	Reverse a string in-place without using built-in reverse functions.	Easy
2	Check if a string is a palindrome — ignore case and non-alphanumeric characters.	Easy
3	Count vowels, consonants, spaces, and digits in a given string in one pass.	Easy
4	Find the first non-repeating character in a string using a frequency map.	Medium
5	Check if two strings are anagrams of each other.	Medium
6	Find the first occurrence of a substring (needle) in a string (haystack) using KMP algorithm.	Medium
7	Find the longest substring without repeating characters.	Medium
8	Perform run-length encoding of a string (e.g. 'aaabbc' → '3a2b1c').	Medium
9	Find the longest palindromic substring using expand-around-centre technique.	Hard
10	Given a pattern (with ? and *) and a string, check if they match using DP.	Hard

Homework Problems

#	Homework Problem	Diff
H W 1	Reverse words in a sentence (not characters) without extra space.	Easy
H W 2	Check if one string is a rotation of another string (use concatenation + substring check).	Medium
H W 3	Find all permutations of a given string (handle duplicates — print unique only).	Medium
H W 4	Group anagrams together from a list of strings.	Medium
H W 5	Find the longest common substring (not subsequence) of two strings.	Hard

Day 6 — Recursion & Recursive Thinking

Instructor Note: For every problem: (1) write the recurrence, (2) identify base case, (3) draw the call tree. Problems 7 & 10 require a full hand-trace.

Classwork Problems

#	Problem	Diff
1	Write a recursive function to find the factorial of N.	Easy
2	Write a recursive function to find the N-th Fibonacci number.	Easy
3	Given base B and exponent E, write a recursive function to compute B^E using fast exponentiation.	Easy
4	Write a recursive function to compute the sum of digits of a number.	Easy
5	Write a recursive function to reverse a given number and return the reversed integer.	Medium
6	Towers of Hanoi: transfer N disks from Source S to Destination D via Auxiliary A, printing each move.	Medium
7	Trace and evaluate: $\text{func}(p,q) = p+q+\text{func}(p-3,q)+\text{func}(p-8,q-2)$ if $p>0$ else $q-2$. Compute $\text{func}(6,27)$ step-by-step and write the call tree.	Medium
8	Given a list of N integers and a target, use recursion to find all pairs that sum to the target.	Medium
9	Josephus Problem: N people in a circle, every K-th person eliminated. Find the survivor's position using recursion.	Hard
10	Trace and evaluate: $\text{mul}(s,k,m) = s/m + k\%5 + \text{mul}(s/3, k/2, m/2)$ if $s+m>1$ else 0. Compute $\text{mul}(12,4,2)$ with a full call trace.	Hard

Homework Problems

#	Homework Problem	Diff
H W 1	Write a recursive function to check if a given string is a palindrome.	Easy
H W 2	Write a recursive function to compute GCD of two numbers (Euclidean Algorithm).	Easy
H W 3	Generate all binary strings of length N using recursion.	Medium
H W 4	Find the power set (all subsets) of a given set of integers using recursion.	Medium
H W 5	Count the number of ways to climb N stairs if you can take 1, 2, or 3 steps at a time, using recursion with memoization.	Medium

Day 7 — Binary Search + Ternary Search + Jump Search

Instructor Note: Binary Search is not just for sorted arrays — apply it on the ANSWER SPACE. Book Allocation and Aggressive Cows are must-solve patterns. All solutions $O(\log n)$.

Classwork Problems

#	Problem	Diff
1	Given a sorted array, find a target value and return its index; return -1 if not found.	Easy
2	Find the first and last position of a target element in a sorted array.	Easy
3	Given a sorted rotated array, find the minimum element.	Easy
4	Search a target in a sorted rotated array (no duplicates).	Medium
5	Find the floor of the square root of N without using <code>sqrt()</code> — use binary search.	Medium
6	Book Allocation Problem: N books with pages, M students — allocate books to minimise the maximum pages assigned to any student.	Medium
7	Aggressive Cows: place C cows in N stalls to maximise the minimum distance between any two cows.	Medium
8	Painter's Partition: K painters must paint N boards — minimise the maximum time taken by any painter.	Medium
9	Given a 2D matrix (rows sorted, first element of each row > last of previous), search a target in $O(\log(m \times n))$.	Hard
10	Median of Two Sorted Arrays of sizes m and n in $O(\log(\min(m, n)))$.	Hard

Homework Problems

#	Homework Problem	Diff
H W 1	Given a sorted array, count how many elements are strictly less than X using <code>lower_bound</code> binary search.	Easy
H W 2	Find the peak element in an array (greater than both neighbours) using binary search.	Medium
H W 3	EKO — Lumberjacks must cut at least M metres of wood; find the maximum saw height using binary search on answer.	Medium
H W 4	Kth Missing Positive Number in a sorted array using binary search.	Medium
H W 5	Find the K-th smallest element in a sorted matrix using binary search on value range.	Hard

Day 8 — Merge Sort & Quick Sort

Instructor Note: No sort implementations — every problem here is solved BY applying the Merge/Quick sort technique. Derive $T(n)=2T(n/2)+O(n)$ for Merge Sort.

Classwork Problems

#	Problem	Diff
1	Count the number of inversions in an array (pairs $iarr[j]$) using modified Merge Sort.	Easy
2	Sort an array of 0s, 1s, and 2s in a single pass without extra space (Dutch National Flag).	Easy
3	Find the K-th largest element in an unsorted array using Quick Select (average $O(n)$).	Easy
4	Given an array, count pairs (i,j) such that $arr[i] > 2 \times arr[j]$ and $i < j$ using modified Merge Sort.	Medium
5	Sort a nearly-sorted (k-sorted) array efficiently — every element is at most K positions from its sorted position.	Medium
6	Find the minimum number of swaps required to sort an array using cycle detection.	Medium
7	Given an array of meeting time intervals [start, end], find the minimum number of conference rooms required.	Medium
8	Given an array, find the smallest range [L, R] such that removing elements outside this range makes the array sorted.	Medium
9	Count pairs in an array whose sum is divisible by K using sorting and two pointers.	Hard
10	Given N rope lengths, find the minimum cost to connect all ropes (always merge two shortest) — solve using sorting.	Hard

Homework Problems

#	Homework Problem	Diff
H W 1	Find the median of an unsorted array in average $O(n)$ time using Quick Select.	Medium
H W 2	Given two sorted arrays of sizes m and n, merge them in $O(m+n)$ using the merge step of Merge Sort.	Easy
H W 3	Find the largest number that can be formed by rearranging digits of an array of non-negative integers.	Medium
H W 4	Find the smallest range that includes at least one element from each of K sorted lists using merge.	Hard
H W 5	Given an array, find the maximum gap between successive elements when the array is sorted — solve without fully sorting.	Hard

Day 9 — Linked List — Part 1 (SLL, DLL, CLL)

Instructor Note: Draw pointer diagrams for every problem. Slow-fast pointer is the key technique for cycle and middle problems. Always handle: empty list, single node.

Classwork Problems

#	Problem	Diff
1	Given a singly linked list, reverse it iteratively and return the new head.	Easy
2	Find the middle node of a singly linked list using the slow-fast pointer technique.	Easy
3	Detect whether a cycle exists in a linked list using Floyd's Cycle Detection.	Easy
4	Find the node where a cycle begins in a linked list.	Medium
5	Merge two sorted singly linked lists into one sorted list.	Medium
6	Remove the N-th node from the end of a singly linked list in a single pass.	Medium
7	Reverse nodes in a singly linked list in groups of K.	Medium
8	Given a doubly linked list, reverse it and verify forward + backward traversal.	Medium
9	Detect whether a given linked list is circular (all nodes eventually loop back).	Hard
10	Given a sorted circular linked list, insert a new value in the correct sorted position.	Hard

Homework Problems

#	Homework Problem	Diff
H W 1	Check if a singly linked list is a palindrome using O(1) extra space.	Medium
H W 2	Find the intersection point of two singly linked lists.	Medium
H W 3	Given a singly linked list, check if it is a palindrome and also find its length — single pass.	Medium
H W 4	Sort a doubly linked list using bubble sort on nodes (swap data, not pointers).	Medium
H W 5	Flatten a multilevel doubly linked list (each node may have a child list).	Hard

Day 10 — Linked List — Part 2 (Advanced Problems)

Instructor Note: LRU Cache is among the most asked questions at Amazon, Google, and Microsoft. Clone with random pointers tests deep copy intuition — draw the three-step pointer diagram.

Classwork Problems

#	Problem	Diff
1	Add two numbers represented as linked lists where digits are stored in reverse order.	Easy
2	Remove all nodes with a given value from a linked list.	Easy
3	Swap nodes in pairs in a linked list without modifying values.	Easy
4	Rotate a linked list to the right by K places.	Medium
5	Reorder list in the pattern $L_0 \rightarrow L_n \rightarrow L_1 \rightarrow L_{n-1} \rightarrow L_2 \rightarrow L_{n-2} \dots$ in-place.	Medium
6	Split a circular linked list into two equal halves.	Medium
7	Remove duplicate nodes from a sorted doubly linked list.	Medium
8	Convert a binary tree (given as level-order input) to a doubly linked list in inorder — in-place.	Medium
9	Clone a linked list with random pointers in $O(n)$ time and $O(1)$ extra space.	Hard
10	Design an LRU Cache using a doubly linked list and a hash map with $O(1)$ get and put.	Hard

Homework Problems

#	Homework Problem	Diff
H W 1	Delete the middle node of a linked list in $O(n)$ without knowing the head count upfront.	Medium
H W 2	Segregate even and odd nodes of a linked list maintaining original relative order.	Medium
H W 3	Given a linked list, check if it forms a cycle of a specific length K.	Medium
H W 4	Reverse a doubly linked list in groups of K nodes.	Hard
H W 5	Design a browser history feature: visit(url), back(steps), forward(steps) using a doubly linked list.	Hard

Day 11 — Stacks & Queues — Part 1

Instructor Note: Monotonic Stack is the go-to pattern for Next Greater/Smaller problems. Deque (double-ended queue) dominates sliding window problems. No data structure implementation questions.

Classwork Problems

#	Problem	Diff
1	Given a string of brackets, check if it is balanced using a stack.	Easy
2	Evaluate a postfix (Reverse Polish Notation) expression.	Easy
3	Convert an infix expression to postfix using a stack.	Easy
4	Find the Next Greater Element for every element in an array using a monotonic stack.	Medium
5	Find the Next Smaller Element to the left of every element in an array.	Medium
6	Given stock prices for N days, find the Stock Span for each day (how many consecutive prior days had price $\leq$ today).	Medium
7	Given a histogram of bar heights, find the area of the largest rectangle that fits inside.	Medium
8	Find the maximum of every sliding window of size K using a deque.	Medium
9	Implement a queue using two stacks with amortized $O(1)$ dequeue.	Hard
10	Find the minimum element in a stack in $O(1)$ time without extra space (modified push logic).	Hard

Homework Problems

#	Homework Problem	Diff
H W 1	Sort a stack using only one additional stack.	Medium
H W 2	Given a sequence of push and pop operations, check if the pop sequence is a valid stack order.	Medium
H W 3	Remove K digits from a number string to get the smallest possible number using a monotonic stack.	Hard
H W 4	Find the largest rectangle of 1s in a binary matrix using the histogram technique.	Hard
H W 5	Given N tasks each with a cooldown interval, find the minimum total CPU intervals to finish all tasks.	Hard

Day 12 — Stacks & Queues (Part 2) + Trees Introduction

Instructor Note: Day split: first half is advanced stack/queue patterns; second half introduces tree problems using BFS and recursion. LCA is among the most asked tree questions.

FIRST HALF — Stacks & Queues Advanced | Classwork

#	Problem	Diff
1	Given a stream of integers arriving one by one, find the running median after each insertion using two heaps.	Medium
2	Decode a string encoded as k[encoded_string] using a stack. E.g. '3[a2[c]]' → 'accaccacc'.	Medium
3	Given arrival and departure times of trains, find the minimum number of platforms needed at a station.	Medium
4	Trapping Rain Water: given elevation heights, find total water trapped between bars using a stack.	Medium
5	Sliding Window Minimum: find the minimum in every window of size K using a monotonic deque.	Hard

FIRST HALF — Homework

#	Homework Problem	Diff
H W 1	Design a circular deque that supports insertFront, insertLast, deleteFront, deleteLast, getFront, getRear in O(1).	Hard
H W 2	Given N intervals, merge all overlapping ones and return the count of distinct merged intervals.	Medium

SECOND HALF — Trees Introduction | Classwork

#	Problem	Diff
1	Level-order traversal (BFS) of a binary tree — print each level on a new line.	Easy
2	Find the height (maximum depth) of a binary tree.	Easy
3	Check if a binary tree is a valid Binary Search Tree.	Medium
4	Find the Lowest Common Ancestor (LCA) of two nodes in a BST.	Medium
5	Check if two binary trees are mirror images (symmetric) of each other.	Medium

SECOND HALF — Homework

#	Homework Problem	Diff
H W 1	Find the diameter of a binary tree (longest path between any two nodes).	Medium
H W 2	Serialize and deserialize a binary tree to/from a string.	Hard

Day 13 — Graphs — Part 1 (BFS, DFS, Cycle Detection)

Instructor Note: BFS uses a queue; DFS uses recursion or explicit stack. Both are $O(V+E)$. Always clarify: directed vs undirected, weighted vs unweighted.

Classwork Problems

#	Problem	Diff
1	Given an unweighted undirected graph, find the shortest path (minimum edges) from source S to destination D using BFS.	Easy
2	Given a grid of 0s (water) and 1s (land), count the number of islands using DFS.	Easy
3	Given a grid, find the size of the largest island (connected component of 1s).	Easy
4	Detect a cycle in an undirected graph using BFS.	Medium
5	Detect a cycle in a directed graph using DFS (back edge detection).	Medium
6	Find the number of connected components in an undirected graph.	Medium
7	Check if a graph is bipartite using BFS (2-colouring — can it be coloured with 2 colours with no adjacent same colour).	Medium

Homework Problems

#	Homework Problem	Diff
H W 1	Find the number of provinces (groups of directly or indirectly connected cities) in an adjacency matrix.	Medium
H W 2	Given a binary grid, find the shortest path from top-left to bottom-right moving 8 directions.	Medium
H W 3	Implement topological sort of a DAG using Kahn's Algorithm (BFS-based indegree method).	Hard

Day 14 — Graphs — Part 2 (Shortest Paths, MST, Advanced)

Instructor Note: Dijkstra: non-negative weights only. Bellman-Ford: handles negative weights, detects negative cycles. Kruskal/Prim: MST — know when to use each.

Classwork Problems

#	Problem	Diff
1	Network Delay Time: find the minimum time for a signal to reach all nodes from node K using Dijkstra.	Medium
2	Cheapest Flights Within K Stops: find minimum cost from src to dst using at most K stops (Bellman-Ford variant).	Medium
3	Find the minimum cost to connect all cities given weighted edges (MST using Kruskal's with Union-Find).	Medium
4	Find the minimum spanning tree cost using Prim's algorithm.	Medium
5	Given a list of courses with prerequisites, return a valid course order (Topological Sort of DAG).	Medium
6	Find all critical connections (bridges) in a network — removing them disconnects the graph.	Hard
7	Find all Strongly Connected Components (SCCs) using Kosaraju's two-pass DFS algorithm.	Hard

Homework Problems

#	Homework Problem	Diff
H W 1	Find the shortest path between all pairs of vertices using Floyd-Warshall Algorithm.	Hard
H W 2	Swim in Rising Water: find the minimum time to reach bottom-right from top-left as water rises (binary search + BFS or Dijkstra).	Hard
H W 3	Find the minimum number of steps to make all cells in a grid reachable by changing some -1 cells using BFS.	Hard

Day 15 — Backtracking

Instructor Note: Template: choose → explore → unchoose. Pruning early is critical. Time complexity is exponential — explain why. N-Queens and Sudoku are top interview questions across all product companies.

Classwork Problems

#	Problem	Diff
1	Generate all permutations of a given array of distinct integers.	Easy
2	Generate the power set (all subsets) of a given array.	Easy
3	Find all combinations of numbers from 1 to N that sum to a target (no repetition allowed).	Easy
4	Find all unique combinations from an array with duplicates that sum to a target.	Medium
5	Generate all valid combinations of N pairs of balanced parentheses.	Medium
6	Given a 2D board of characters, check if a target word exists by traversing adjacent cells.	Medium
7	Letter Combinations of a Phone Number: given digits 2-9, return all possible letter combinations.	Medium
8	N-Queens: place N queens on an N×N board such that no two queens attack each other — return all configurations.	Medium
9	Solve a partially filled 9×9 Sudoku board.	Hard
10	Rat in a Maze: find all paths from (0,0) to (N-1,N-1) in a binary matrix (move U/D/L/R).	Hard

Homework Problems

#	Homework Problem	Diff
H W 1	Generate all permutations of a string with duplicate characters — return only unique permutations.	Medium
H W 2	Combination Sum III: find all combinations of exactly K numbers (digits 1-9, no repeat) that sum to N.	Medium
H W 3	Palindrome Partitioning: partition a string so every substring is a palindrome — return all valid partitions.	Medium
H W 4	Knight's Tour: find a sequence of moves so a knight visits every cell of an N×N board exactly once.	Hard
H W 5	Expression Add Operators: given a digit string and a target, insert +, -, * between digits to reach the target.	Hard

Day 16 — Greedy Algorithms

Instructor Note: Greedy works when a locally optimal choice leads to globally optimal. Always prove your greedy with an exchange argument. Show where greedy fails for Coin Change vs DP.

Classwork Problems

#	Problem	Diff
1	Activity Selection: given start and finish times of N activities, select the maximum number of non-overlapping activities.	Easy
2	Fractional Knapsack: maximise total value for a given weight capacity — items can be taken partially.	Easy
3	Minimum number of coins to make amount N (greedy works for standard denominations — prove when it doesn't).	Easy
4	Job Sequencing with Deadlines: given jobs with profit and deadline, schedule jobs to maximise total profit.	Medium
5	Huffman Encoding: given character frequencies, build a Huffman tree and print codes for each character.	Medium
6	Jump Game: given array of max jump lengths, determine if you can reach the last index.	Medium
7	Jump Game II: find the minimum number of jumps to reach the last index.	Medium
8	Minimum number of platforms needed at a railway station given arrival and departure times.	Medium
9	Assign Cookies: given children's greed factors and cookie sizes, satisfy as many children as possible with minimum cookies.	Hard
10	Reorganise String: rearrange characters so no two adjacent characters are the same — use greedy with a max-heap.	Hard

Homework Problems

#	Homework Problem	Diff
H W 1	Gas Station: find the starting index from which a car can complete a full circular route given gas and cost arrays.	Medium
H W 2	Candy Distribution: give children candies such that children with higher ratings get more than their neighbours — minimise total.	Hard
H W 3	Minimum number of arrows to burst all balloons on a 2D plane.	Medium
H W 4	Task Scheduler: find minimum CPU intervals to finish all tasks given a cooldown constraint.	Medium
H W 5	Partition Labels: partition a string into as many parts as possible so each letter appears in at most one part.	Hard

Day 17 — Dynamic Programming — Part 1

Instructor Note: Three-step approach: (1) define subproblem, (2) write recurrence, (3) identify base case. Implement both memoization and tabulation for each problem.

Classwork Problems

#	Problem	Diff
1	Fibonacci using DP — compare recursive, memoized (top-down), and tabulated (bottom-up) versions.	Easy
2	Climbing Stairs: count the number of ways to reach the N-th stair taking 1 or 2 steps at a time.	Easy
3	0/1 Knapsack: given weights and values of N items and capacity W, maximise value without fractional items.	Easy
4	Longest Common Subsequence (LCS) of two strings.	Medium
5	Longest Increasing Subsequence (LIS) — solve in $O(n^2)$ DP and then $O(n \log n)$ using patience sorting.	Medium
6	Coin Change (minimum coins): find the fewest coins to make amount N — show where greedy fails.	Medium

Homework Problems

#	Homework Problem	Diff
H W 1	Unique Paths: count distinct paths from top-left to bottom-right of an $m \times n$ grid (only right/down moves).	Easy
H W 2	Partition Equal Subset Sum: can the array be split into two subsets with equal sum?	Medium
H W 3	Edit Distance: minimum operations (insert, delete, replace) to convert string A to string B.	Hard
H W 4	Matrix Chain Multiplication: find the minimum number of scalar multiplications to multiply a chain of matrices.	Hard

Day 18 — Dynamic Programming — Part 2

Instructor Note: Interval DP (Burst Balloons, Matrix Chain) and State Machine DP (Stock Problems) are advanced patterns. Day 18 also serves as a full DP pattern revision.

Classwork Problems

#	Problem	Diff
1	Word Break: given a string and a dictionary, check if the string can be segmented into dictionary words.	Easy
2	Palindromic Substrings: count all palindromic substrings in a string.	Easy
3	Rod Cutting: given a rod of length N and prices for each length, maximise revenue by making cuts.	Easy
4	Maximal Square: find the side length of the largest square containing only 1s in a binary matrix.	Medium
5	Burst Balloons: burst balloons in an optimal order to maximise total coins collected.	Medium
6	Best Time to Buy and Sell Stock with Cooldown: maximise profit with one-day cooldown after selling.	Medium

Homework Problems

#	Homework Problem	Diff
H W 1	Egg Drop Problem: find the minimum number of trials to determine the critical floor from N floors with K eggs.	Hard
H W 2	Wildcard Matching: implement * (matches any sequence) and ? (matches one character) using DP.	Hard
H W 3	Minimum Cost to Cut a Stick: given cut positions on a stick, minimise total cutting cost using interval DP.	Hard
H W 4	Regular Expression Matching: implement . and * regex matching using DP.	Hard

18-Day Curriculum Overview

Day	Topic	CW	HW	Difficulty Split
1	Bit Manipulation	10	5	$3E \cdot 5M \cdot 2H$
2	Number-Based Problems	10	5	$3E \cdot 5M \cdot 2H$
3	Pattern-Based Problems	10	5	$3E \cdot 5M \cdot 2H$
4	Array Problems	10	5	$3E \cdot 5M \cdot 2H$
5	String-Based Problems	10	5	$3E \cdot 5M \cdot 2H$
6	Recursion	10	5	$4E \cdot 4M \cdot 2H$
7	Binary Search + Ternary + Jump	10	5	$3E \cdot 5M \cdot 2H$
8	Merge Sort & Quick Sort	10	5	$3E \cdot 5M \cdot 2H$
9	Linked List Part 1	10	5	$3E \cdot 5M \cdot 2H$
10	Linked List Part 2	10	5	$3E \cdot 5M \cdot 2H$
11	Stacks & Queues Part 1	10	5	$3E \cdot 5M \cdot 2H$
12	Stacks+Queues Part 2 + Trees	10	4	$2E \cdot 6M \cdot 2H$
13	Graphs: BFS, DFS, Cycles	7	3	$3E \cdot 3M \cdot 1H$
14	Graphs: Shortest Path, MST	7	3	$0E \cdot 5M \cdot 2H$
15	Backtracking	10	5	$3E \cdot 5M \cdot 2H$
16	Greedy Algorithms	10	5	$3E \cdot 5M \cdot 2H$
17	Dynamic Programming Part 1	6	4	$3E \cdot 2M \cdot 1H$
18	Dynamic Programming Part 2	6	4	$3E \cdot 2M \cdot 1H$